Two cases on how to improve the visibility of business process performance

Tim Pidun
Technische Universität Bergakademie Freiberg
Chair of Information Science
tim.pidun@bwl.tu-freiberg.de

Carsten Felden
Technische Universität Bergakademie Freiberg
Chair of Information Science
carsten.felden@bwl.tu-freiberg.de

Abstract

Performance measurement of business processes is necessary to ensure adequate information supply to the management. Though, the assessment of rather qualitative or non-deterministic processes is hard, because common analyzing and controlling approaches are restricted to the use of numeric indicators. Important qualitative performance aspects remain invisible. To bridge this gap, we introduce the concept of visibility which defines appropriate information supply in the domain of process performance. Derived from these requirements, we also present a multi-dimensional performance assessment system that combines numeric and verbal indicators. In two cases, we prove its applicability and positive contribution to performance visibility.

1. Introduction

Business Process Management features three basic functions: process organization, process optimization and process controlling [1]. The analysis of performance by Performance Measurement Systems (PMS) is usually located in the domain of controlling [ibid]. Empirical studies reveal a gap between process analysis, implementation, and execution [2], [3] as well as a gap between controlling and process management [1]. One reason is that business process performance outputs are usually supposed to be measured and counted [43], [44]. Other qualitative and rather descriptive performance aspects of process performance that are well known in process management like complexity, probability of failure, goals, roles or relations cannot easily be assessed in this way. Hence, some performance aspects are less *visible* than others that cannot easily be measured [ibid]. With performance being also defined as the degree of stakeholder satisfaction [8], satisfaction can only be tested if performance is *visible* to the stakeholders at all.

The goal of this paper is to introduce the concept of *visibility* as key factor to reveal information on process performance that is hard to catch by common process controlling tools (PMS).

PMS rely mostly on numeric oriented key performance indicators (KPI) [14], [17] which are strongly related to highly specified, formalized, and automated business processes [11]. Especially support processes like Human Resources or Public Relations that are related to qualitative outcomes are hard to measure [20], [19], [23] and hence are lost for review and control through KPIs. Being aware of this fact, some PMS try to consider qualitative key figures to measure processes. E.g. the Performance Prism of Neely et al. [13] considers strategy, capabilities, and stakeholder views. Kaplan and Norton [12] have introduced the Balanced Score Card (BSC) including different perspectives like finance, customer, development potential and process perspective. All these PMS focus on comprised qualitative and quantitative KPIs. An assessment of descriptive process success factors is not enabled and complex interdependencies, human interactions and capabilities are not sufficiently represented. Alternatively, there are other multi-dimensional approaches that incorporate goals, roles and relations [23], [40]. Especially the Four-Box *Performance Assessment System* (PAS) [4] features four different viewpoints to performance problems and broadens assessment possibilities from numeric KPIs to three additional generic indicators.

The contribution of this paper is a procedural model to apply this *PAS* as well as two case studies in which it was applied. Using the PAS, invisible performance can better be detected and used as entry points for improvement. We constitute an Operationalization of this system and establish a proof of concept of *visibility*.

The remainder of this paper is as follows: Section 2 explains the problem statement with status quo, theoretical background and approach, leading to the research questions. Section 3 depicts the research design and the settings of the case study. Section 4 consists of the investigation of the two cases and the findings related to the application of the model. The final section summarizes the results, explains the benefits of the contribution and gives a future outlook.

978-0-7695-4525-7/12 $26.00 © 2012 IEEE
DOI 10.1109/HICSS.2012.608

2. Problem statement

Improving the performance of business activities requires the right information to identify, analyze, and redesign business processes. Getting the right information is essential to reduce uncertainty and take appropriate decisions [5]. Measurement is the key to give the management information, helping to reduce uncertainty and to take appropriate decisions [6]; one of the key demands of Business Process Management (BPM) is that performance of processes has to be measured (e.g. [1], [7], [8]).

But performance measurement is not as successful as it may seem [7], though many recommendations, methods or software solutions exist (compiled e.g. by [10], [11]). At the same time, empirical studies reveal a gap between process analysis, implementation, and execution [2]. The need to conduct BPM with alignment to strategy is also emphasized by various empirical studies, e.g. [14] but only a minority of the surveyed companies sees a correspondence between strategy and BPM [3]. Strong reasons are e.g. the lack of methodological guidance of PMS due to a strong orientation to *numeric* indicators [16], [17], as well as a bad *visibility* of performance in specific qualitative areas [14], [43]. These two aspects are being discussed in the following sections.

Measurement replaces trust and confidence with compliance and assurance [5]. This paradigm has pervaded business domains and organizations that were not in focus of audit and control before [18], like public administration or health care. Processes that feature complex qualitative performance problems are reduced to single parameters [19] and circumscribed with various range of measures [20], [21], [22]. Extra cost for finding, defining, applying, and maintaining measures [20], [23], [24] often lead to an ignorance of qualitative measures [25]. Other indirect formalized concepts like complexity, maturity, goals, roles or relations that relate to indicators [40] are underrepresented in Performance Measurement [26]. If processes with qualitative backgrounds are not assessed at all or are hard to catch via circumscribing measures, performance problems stay invisible. The result is a reduced comprehension and *visibility* of performance that prevents the delivery of valuable information and consolidates the gap between strategy and business processes.

Recent empirical studies originating in the domain of *performance management* introduce the term *visibility* in the sense of *quality of information* [14], [43]. It depicts the degree of comprehension of business process performance with the help of appropriate indicators and fosters a certain *oversight* on performance *information*.
In addition, there is a similar concept of visibility emerging from the domain of *supply chain management* [28], [45], [41]. There, *visibility* is defined as the ability, usefulness, usability, timeliness, trust, and accuracy to exchange *information* between supply chain partners. Benefits of good visibility are e.g. total distribution cost, a lower inventory level, holding-, shortage- and total cost, higher delivery-, firm- and market performance as well as flexibility and quality. Visibility can be established by using an appropriate supply chain management *system* [28]. The supply chain is usually considered to be a fundamental business process. Other researchers assume that the multitude of business processes can also be seen as internal supply chains as well and hence the two domains are comparable to a large extent [15], [42]. The additional fact that the success of the Supply-Chain Operations Reference (SCOR) model as popular assessment method for supply chains is also driving its distribution in other business process domains like product development, CRM, finance etc. [42] is another complementing commonality between the SC and business processes in general. Finally, they also support the analogous use of visibility to describe phenomena on business processes in general and in particular in performance assessment. So, both approaches describe the need to deliver increased *information supply* for the *management*.

Consequently, we have adapted this concept and defined *visibility* of *process performance* as the principal ability to assess process performance *information* [27]. It is established by using a suitable performance measurement system that delivers appropriate information to the management. A PMS is designed on the basis of a three-step-procedural model according to the original approach described in [28]:

1. Classification of the main types of information, which can be exchanged.
2. Definition of the most important features of the information flows.
3. Development of a set of metrics and a measurement model to assess the level of visibility.

To increase this visibility of process performance, we have developed the *Four-Box* PAS as assessment model in previous work [4]. It combines aspects of measurement and analysis that both originate in process controlling. It features four different systematic viewpoints (so-called *generic indicators*) to business performance problems and broadens assessment possibilities. Due to this interlaced design, the model incorporates all business processes and covers all domains in principle, not only production or value creating processes. It is designed to be useful especially in domains or processes that are hard to analyze and to describe by numeric parameters.

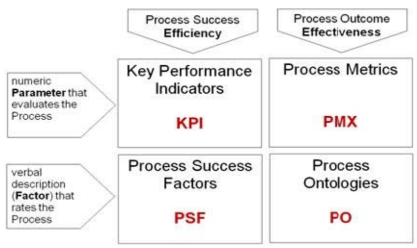

Fig. 1 The Four-Box model and PAS

To form the *generic indicator*s, we collected interrelated formalized concepts that point to a potential use as indicators and abstracted their working principles in two dimensions. First concerning *efficacy,* they assess efficiency through evaluation of process success as well as effectiveness through process outcome. This underlines the common perception of performance, which describes both the output (success) as well as the ability to perform (outcome) [1]. Within a second dimension, *embodiment*, they either assess the process performance by the assignment of a certain numeric value as usual, or additionally by verbal description and judgment, thus avoiding the common categorical quantization of performance.
In detail, the generic indicators (boxes) consist either of interrelated numeric parameters that point to process success, e.g. measures like amount, throughput, cost etc. (KPI), verbal hard or soft goals, success factors (PSF), numeric parameters pointing to internal quality like complexity, knowledge, probability of failure, cycle time etc. (PMX) as well as semantics, e.g. ontologies, semantic annotation or role and relation related information (PO) [4]. On iteratively analyzing the specific performance problem from every one of four viewpoints, real and meaningful inherent *performance indicators* contained in the process can be determined.

This investigation fosters the research question whether the PAS is able to enhance the visibility of process performance in principle, and based on these considerations, whether this concept of visibility is able to support knowledge and understanding of process information, problems and performance in order to finally improve the correspondence between process management and controlling and strategic level. To describe and verify the applicability and usability of the PAS, it was applied to two cases of process optimization in the rather supportive domain of procurement on the processes for Capital Expense (CAPEX). This also embodies an Operationalization of the PAS construct.

3. Research Design

The research design orientates on the Design Science approach of Hevner et al. [29]. The status quo as basis for the investigation has already been explicated in Section 2. The investigation approach and body is described in this Section. First, we explain the chosen qualitative method (case study) along with framing aspects of triangulation, replication, the case settings, followed by the description of the application of the PAS by using a procedural model to develop suitable performance indicators for the the two cases of process improvement. The actual investigation course is described in Section 4. Results and implications of the study as well as reflections on usability and practical advantages in Section 5 frame the investigation which can be classified as *research-in-progress* since this contribution is just one first possible validation of the concept of visibility. Details are described in the concluding paragraph on future research.

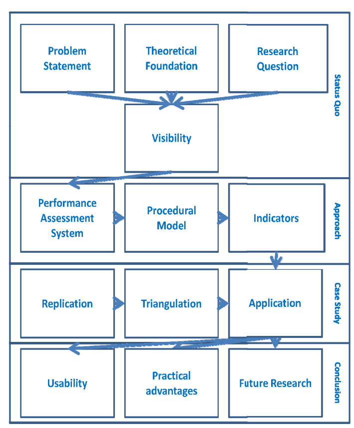

Fig. 2 Research Design

This investigation aims at BPM and the visibility of business process performance. Porter [30] differentiates processes in those that are primary value creating (part of a value chain) and supportive processes like infrastructure (e.g. finance, public relations) human resources, technical development as well as procurement. In contrary, Rüegg-Stürm [31] differentiates between three process categories that are all value-adding in their perception: management processes (e.g. organizational development, steering, control), business processes (creating immediately customer value) and support processes (infrastructure and internal services). Besides that, Zellner [32] defines business processes in general as processes that describe the essential functions of an enterprise and which execution contributes to its success altogether. These three viewpoints have in common that supportive processes are not to be omitted in optimization projects. Though opinions vary concerning their level of adding value and facilitation of the company overall success, in principle their contribution is undoubted. CAPEX in particular can be considered as an important part of the company procurement activities that in turn are part of the support activities of the business process ecosystem according to Porter. Summing up, the chosen process is valid as body of investigation, especially in the context of its rather qualitative nature and the difficulty to assess it with numeric values.

According to Yin [33], a case study is "an empirical inquiry that investigates a contemporary phenomenon within its real-life context, especially when the boundaries between phenomenon and context are not clearly evident". We chose the case study since the situations in which the research questions could be raised and processed were rather incidental, ad-hoc and not experimental in the sense of the ability to plan the settings and occurrence before. Moreover, the case study as such enables to quickly gather much qualitative information on the case that can be extended and investigated in detail later on.

In case one, the body of investigation could be observed simply, because of having had the opportunity. One of the main stakeholders of the CAPEX process of this company was the department in which we were accountable for the respective improvement endeavors from March 2009 to March 2010. We were invited to contribute to the optimization of this super ordinate process, too, as the common interfaces and interests of improvement became obvious. So we performed supplemental interviews and discussions with the CAPEX process owners, stakeholders and customers. In this case, we held the role of (additional and passive) external observers and advisors to the Lean Management team without being involved in the optimization itself. In case two, the CAPEX process was the first to be considered in a row of project workshops in a public company starting January 2011 with the final goal to define the necessary specifications for central software deployment. Prior to the description of the requirements, an optimization of the processes had to be performed. In this case, we actually acted actively as the analysis and improvement team. In contrary to case one, in which the workflow was rather organically grown, the CAPEX process case two was strictly aligned to the local legal requirements. Summing up, the two cases featured a similar process success but extremely different approaches and outcome.

This ability to use other and different cases for the investigation of the same problem is called replication and considered to be a key necessity of a scientific validation in principle [34]. In particular the case study method should be offering a bundle of data sources to adequately justify its validity through the method of triangulation [35]. Yin refers to six possible data sources [33]: documents, archival records, interviews, direct observation, participant-observation and physical artifacts. Another possibility of consensual validation is peer consultation. The used methods as well as commonalities and differences of the two considered cases are summarized in Table 1.

Aspect	**Case one**	**Case two**
Process	CAPEX with focus on equipment acquisition	
Ownership	Private	Public
Branch	Semiconductors	Education
Employees	2500	200
Customers	20	5000
BPM Team	2	2
Process Responsible	Manager of Lean Manufacturing	Dedicated process responsible
Available documentation	BPMN Notation	ARIS-EPC Notation
Further triangulation by	Interviews with process responsible	Written description and directives
	Peer consultation with other process stakeholders	Workshop with process responsible
Basic optimization approach	Lean manufacturing	Six sigma

Table 1: Overview of case settings

The main differences of the two cases are the ownership, branch, size, and customer base of the hosting companies that are virtually diametrically opposed. A notable fact is the same small size of the BPM team being in charge of the process optimization independent of the size and scope of the project. Process notations existed in both cases and some other data sources that could be used for triangulation were available. The main common problem on assessing these processes prior to the application of the Four-Box model was the use of relatively unsuitable basic optimization approaches because of their reliance on numeric values. Both approaches originate in business process *analysis* and are well-known in process optimization. The effects of Lean manufacturing e.g. are being measured in pieces of work in progress (WIP), cycle time, cost, quality and revenue aligned on targets and milestones [36]. The six sigma methodology uses measurement in step two of the define, measure, analyze, improve, control and report (DMAICR) method [37]. Both approaches are based on the idea of continuous improvement [38] with measurement hidden in step three of the well-known Deming cycle: plan, do, check and act. Moreover, Deming states that mechanisms of improvement can better be visible through statistics [39], thus pointing to common statistical process control (SPC) methods which of course presume numeric values.

Because of the nature of a supporting procurement process in principle and the CAPEX process in particular, it is not in focus e.g. how many instances are ran to gain statistical significance, how much equipment exactly is purchased or how quickly decisions are made as long as the procurement supports the enterprises' goals adequately by means of e.g. budget and compliance. The success of the CAPEX process and its value for the enterprise can only be existing or not, but it rarely can be assessed with a certain parameter nor named to be good or bad. Hence, the optimization approaches that aim to optimize the process success were not helpful. The goal of the reengineering work not necessarily needs be put down to better results, but to a smoother execution with focus on the process outcome as a whole. The success of the process (e.g. execution in time or compliance) might usually be independent of the fact that e.g. either five or fifteen employees were necessary to archive it, or if two or six versions of a scenario were used to find the final contribution. Alternatively, the Four-Box model as an assessment system can be used to describe the process outcome. It originates in performance measurement and hence in the business process *controlling* domain.

To operationalize and apply the Four-Box-model and collect the case data, we were acting according following *procedural model*: During the interview or workshop with the responsible, we *collected* key aspects of the verbal process descriptions and a rough graphical workflow notation in free form on flip-charts (case one) and on semi-structured questionnaires (case two). Already while and after the collection, we tried to *detect* and extract the inherent performance indicators of the processes by asking questions aiming to the contents of the four generic indicators: In case the success or the outcome of a process could not easily be described by KPIs, we determined the scope and goal of the assessment and either tried to find performance indicators for the description of success, or the features that contribute to success, belonging to the class of PSF. Alternatively, we were seeking indicators that matched the class of PMX for the evaluation of intrinsic outcome features as next successive indicator. Wherever possible, framing links to roles, relations and interdependencies of building blocks or stakeholders of the process were collected, thus serving the generic indicator PO. In detail, the following questions were asked:

Generic Indicator	Question
On KPI	1. What Key Performance Indicators are used or visible throughout the process?
	2. If not explicit, are there specific countable amounts or ratios, cycle time or cost of products or services to be delivered or achieved by the process?
On PMX	3. Are there numeric parameters linked to quality, e.g. excursion ratio, probability of failure or loopback or media interruptions?
	4. Is a certain maturity of a process, complexity of a task or necessary knowledge of the responsible visible or describable?
	5. Can these indicators be compared to other occurrences, the past, similar elements or be rated on a scale?
On PSF	6. Why is this process in place? What are the scope and benefit of this process and the steps?
	7. Having achieved what to what extent, the process or the step could be called a success?
	8. What information systems, databases and interfaces are being used?
	9. What documents and information are produced and delivered as a result of the process and to what extent are they useful?
On PO	10. Who are initiators, beneficiary and customers of the process?
	11. Who are the responsible, who are the approvers of the step or process?
	12. Is there a key person or information system for a step or is there shared knowledge or backups?
	13. What departments, instances or other third parties are affected by or dependent on the process?

Table 2: Investigation questions

After the interviews, the results were condensed and the workflows re-notated. By *interpreting* the results of this assessment, two goals can be served. First, entry points to improvement can be identified (which originally was the main goal of the workshops). Second, by comparing the found indicators, their combination and derivatives, conclusions on the focus, relative efficiency and effectiveness of the processes and tasks could be also made between the two cases. Finally, we recorded the subjective *perception* of a change in process performance information plurality and quality which finally leads to a change of *visibility.*

4. Investigation

The investigation initially features the actual description of the two cases. Their first paragraphs refer to the basic description of scope and surroundings of the process optimization, the following to the documentation and data sources that were of use for the triangulation. The following section contains the results of the analysis according to the Four-Box model follows, explicated in a table sorted by the four generic indicators. Finally, an interpretation of the respective case analysis and a short summary of the results and implications are given.

4.1. Case One

The CAPEX process of a large multinational semiconductor manufacturing company is investigated. The process of a big local site was taken as blueprint for the alignment of the other companywide processes on equipment planning and purchasing to find a single leading corporate process.

The process was used to calculate demand, production capacities and hence needed equipment for a given period and to ensure both sufficient production capability and appropriate ROI of the assets. Unfortunately, there was no written documentation concerning need or objective nor a description of the roles and responsibilities between the parties in a sense of quality relevant documentation.

We were supplied with the BPMN notation of the process. It was claimed to have been developed rather ad-hoc by the staff, as no such notation had existed explicitly and BPMN knowledge was not present in the department before.

The process was modeled by using swim lanes for different departments and directorates and contained not only regular symbols and a process flow but many free comments to the elements that indicated a certain inherent aggregation of explanations, conditions and constraints. Some process elements were commented with respect to a periodic occurrence, e.g. monthly or quarterly approval meetings, some referred to inputs and outputs unquoted in the actual notation, some to additional sub- and external processes. In some cases, the step ownership given through the swim lane was not the same than in the explanation or notation, so that we had to ask for clarification before diagnosing. All in all, we had to re-notate the process in order to structure the workflow, loops, dependencies and external relations. The notation was redone in ARIS-EPC that explicitly featured the data and role resources that had somehow been mixed into the descriptions before.

The process originally was designed for maximum security on order to avoid costly misinvestments. These costs were considered to be an indicator for the performance of the process. The interview partner in case one described the process being too *overburdened*. He put it to the point when claiming that *already three senior site executives found optimizing this process a tough nut to crack*. Due to the multitude of possible exceptions and co-determination possibilities through the lifecycle phases of the process, the “normal” process would never be used. The Lean manufacturing optimization approach common in the company could not be used efficiently, because the waste in the sense of Kaizen’s *muda* could hardly be put down to figures. Moreover, this process was rich of review cycles and interdependencies without clear cause-effect relations and responsible. So the perception of the problem was that the process was too complex and poorly managed.

Other stakeholders of the process even perceived it as *unserviceable anyway*. They followed their own workflows for some parts which were covered by their own process descriptions in the form of as text processing documents on the technical details or implicit knowledge of the stakeholders. Sometimes, even the data bases that lead to the analyses and reports were questioned and tried to be disproved by other participants of the approval sessions (“but my data shows that…”). This indicated a certain lack of ownership and commitment to the previous delivering departments. Some amended the thwarting fact that the processors of some tasks were separated to the approvers by means of location and time zones as a result of the multinational embodiment of the company structure, a fact that cannot be avoided, but must be encountered with synchronization. Some other departments had only one single key person in charge to deal with the requirements of the process.

The first analysis run revealed that the huge process actually consisted roughly of four relatively independent parts that delivered input for the other parts in a row. We identified following sub-processes that could have been subject to individual reorganization and -optimization themselves: Demand planning, collation of an equipment list, approval through the instances, budgeting with quotation and finally preparation of a purchase order. At least, partial responsibility from input to output should be established.

Another characteristic of the process that we noticed was the multitude of periodic activities without visible deadlines or instructions for deliverables and approval. We also noticed the occurrence of many reviewing sessions with many approvers compared to the number of basically responsible, mainly because of the wish to have everyone engaged before the final decision. They in turn resulted in many review loops; sometimes back close to the process start. In conclusion, there have never been scenarios that passed the gate to the next process section at once; there were always changes and reviews before approval. Though, these facts were invisible to the primarily introduced KPIs. So, we recommended introducing some deadline-driven approved results that are no longer questioned (so-called plans of record, POR) that should be strictly synchronized to the appropriate committee occurrences.

Our perception was that this constellation would reduce many additional approval loops, different versions and lifetimes of PORs and hence typically in delays in the numerous affected directorates.

The proposal to introduce specified common workflow rules and clear responsibilities helped to oblige the given process to some extent, but there was no direct attempt to change it by the actual reengineering team while we additionally analyzed the process.

4.2. Case Two

The CAPEX process is one of the most important and integral parts of the company workflow that still lacks support by information systems and is the first in a row of processes that had to be analyzed and optimized in order to set up a final software specification. As most of the information transport through the process was based on paperwork, copies and physical filing, statuses, and implications for other related departments in any phase of the process were very hard to catch. A third-party central purchasing portal that offered the main and common activities of the process for other public companies as well (supplied by the federal authorities) was in discussion but not funded yet. One of the main goals of the process was to be in compliance with regulations and laws that apply to public administration (esp. public procurement law) and to the principal command of transparence to the public. Though the stakeholders perceived the process as working well in this context, they had the notion of *something* would be suboptimal because of many *hiccups* they had experienced while daily work.

The process itself was already pre-visualized by the responsible department in an ARIS-EPC. Additionally, there was a written documentation (procurement directive) describing regulations, tasks, and involved departments available as text file.

From a historical point of view, main portions of the process were broken up into different responsibilities and workloads for departmental and budgetary reasons which did not reflect the overall process driven approach. Though, backups for every process step were available.

The workshops themselves were divided into two parts: the compilation of the actual condition and development and analysis of the target state. Already while project conceptualization we agreed to use a Lean Management approach to detect unnecessary *waste* in the process as entry point for the problem description. Primarily, the evaluation of the relations and interfaces between the stakeholders of the process as well as the implicit and undocumented repositories and databases were supposed to be helpful to contain valuable insights.

One of the key results of the workshops was the implicit notion to make the results of the process *airtight* to internal and ministerial revisions and external complaints. The goal was to spend the public money appropriate and comprehensible. Moreover, the alignment to the accounting and reporting deadlines was of interest.

This is especially hard in the case of premature orders from the senior managers that exceeded the authority limit (approximately 25% of all cases) that had to be justified and financed with retrospective aspect and many additional efforts and explanations. Possible, but harsh solutions are sanctions for the releasers manly through the consideration as *private* orders that cannot be refinanced. The communication of the validity and compliance to the procurement directive was to be enhanced.

In the phase of final procurement, there was no central and consistent supplier database, but only an unsupervised list of possible payees that contained many double entries with different syntax in company names, departments, addresses and contact persons. The same was true for inconclusive payments that made up about 5% of the transactions, mainly because of a missing reference. This delayed or looped payments quite often despite the expertise of the pay office. A solution for these problems would be the mandatory and central assignment of case numbers independent of their size that unambiguously identify the project and avoid separate and uncoordinated orders. In addition to that, the participants proposed to establish a central invoice receipt center that avoids wrong cost assignment through wrong recipient responsibility

On large orders, the early involvement of supporting departments and facilities was not given, so that in one worst case, equipment was delivered and financed in time (due to a valid contract), but without the building and media being ready yet because of the missing request for their funding. A central fixed assets department or administrative software that would be able to administer these acquisitions including trailed facilities and supplementations would be helpful in this case.

4.3. Results

In this section, the results of the application of the Four-Box PAS to the two cases are collected and discussed. The performance indicators that could have been identified while analysis depict how, and how well the process is working.

	Performance Indicators	
Generic Indicator	**Case one**	**Case two**
KPI	ROI	n/a
	Cost of misinvestment	
PMX	Cycle time of approval meetings	Rate of inconclusive payments
	Amount of approval meetings	Rate of exceptions from the procurement authorization limit
	Amount of approvers	
	Amount of responsibles	
	Probability of exceptions	
	Amount of review cycles (plan versions)	
	Probability of reviews	
PSF	Meeting of and alignment to the deadlines	IS- supported access to project data
	Reliance in one single database	Compliance to the laws and regulations
	Generate definite plans-of-record	Transparence to the public
	Establishing of four sub-processes	Reliability in case of revisions
	Delivery of demand, equipment list, quotation and purchase order	Appropriate spending of public money
PO	Distinction of role between responsible or affected/informed departments	Relations to other involved departments
	Role of approvers and responsibles	Interfaces to other departments
	Role of final (sub-) process owners	Relations to other external stakeholders
	Conditions and constraints	Definition of central invoice receipt center
	Location and time zones	
	(informal) Commitment	

Table 3: Case-related performance indicators

4.4. Interpretation

In case one, there are actually conventional KPIs given that are used to measure the success of the process whereas in case two, no direct countable results of the process were eluded. This might be, because of the usual overall application of KPIs to a lot of processes in the industry. They are used to depict their contribution to the financial success of the company. In contrary in case two, many process success factors (PSF) pointing to the overall process success were visible. In case one, they were rather related to single parts or building blocks of the process. Numeric Process Metrics (PMX) could be determined in both cases. Indicators in the first company rather point to detail-related, over managing and organizational issues that lead to poor overall performance whereas in case two, the final rate of exceptions of the process as a whole is the point of entry. Semantic aspects (PO) are found in both cases, mainly related to the explication of existing and desirable roles and relations. In case one, additional constraints are time and location of different stakeholders. Finally, the interesting notion of (subjective) commitment could also be interpreted as indicator of coordinated process execution.

As a result, in case one there were four major optimization approaches discussed, serving the *decentralization* idea: Segmenting of the process with establishment of a respective clear decision arrangement with dedicated ownership, establishment of a single point of truth (database) and introduction of deadline-driven PORs in the respective segments. Nevertheless, the success of the process could still be made measurable with the present KPIs, preferably by a significant change of the cost of misinvestments. In case two, the installation of *central* authorities was the proposed optimization approach of choice. A central database for project proceedings with case numbers and links to the ordering party, suppliers, and payees was demanded to enhance transparence and reproducibility, as well as a (virtual) central invoice receipt center. These approaches were flanked by clear communication of the categorical process ownership of the procurement department and the procurement directive to all stakeholders, as well as sanctions in case of noncompliance.

Up to now, the participants of the interviews and workshops vastly verbally *agreed* to the subjective *usability* and usefulness of the approach as numerous performance problems and waste became *visible* that were hidden in perceived untouchable quantitative processes that therefore, previously remained untouched.

5. Conclusion

In both cases, additional CAPEX process performance indicators pertaining to the four viewpoints of the Four-Box model could be identified and addressed to the process responsibles. Hence, the application of the Four-Box PAS to a prototypal qualitative process was *possible*. The procedural model that was used in the two cases can be considered as a valid Operationalization of the PAS. It gives the management additional valuable *information* on performance and working principles of the business processes, on the one hand through the possible improvement by

1. Conventional improvement by changing figures (e.g. reduction of cost of misinvestments or occurrences of exceptions), suitable for most process optimization approaches that use numeric values, like Lean Management or Six Sigma
2. Goal achievement or approach (e.g. meeting of deadlines or better reliability in case of revisions),
3. Implementation of framing or explaining semantic conditions (e.g. definition of ownership or explication of interdependencies),

on the other hand through the possibility to check their alignment to the strategic goals of the company.

In summary, the PAS combines methods from both process controlling and analysis, bridging the gap between process execution and controlling. Its multi-dimensional combination of numeric and verbal description to assess performance is a new approach, can be applied virtually out-of the box and without extra cost. It overcomes the measurement paradigm that relies on (economics-driven) *number-crunching* and enables the application of other viewpoints. It fosters the *visibility* of business process performance through extending information assessment possibilities. Finally, this concept of *visibility* itself can be considered as a suitable indicator to determine the usefulness and value of this (management) system PAS *to the management*, compared to other PMS.

There are some other similar, but delimiting approaches that can be used to further elaborate the current contribution, first from the view of process *analysis*. The *Process Performance Measurement System* (PPM) e.g. [23] also considers qualitative aspects of performance through the definition of descriptive goals and recommends the use of secondary scores or indicators that state to what extent the goals are fulfilled, but do not feature a role-related viewpoint. The concept of the generic *Performance Indicators* (PI) [40] also implements qualitative as well as quantitative performance values and adds some more characteristics like roles, goals and relations to other performance indicators or processes in an *own* generic modeling language. Second, process *controlling* and measurement approaches like most *PMS* of course are able to assess performance in detail in comparison, but strictly rely on numeric evaluation [26]. Well-established numbers-driven process optimization tools like Six Sigma or Benchmarking are closely interrelated to them, but follow the same restriction.

These comparisons lead to the discussion of the limitations of this contribution. First, the definition of visibility remains strictly conceptual in this paper. In this state, it can only be proven that it is applicable to compare two assessment approaches for a specific case in order to state that one is performing *better,* i.e. that there is more and appropriate information. Though this goal might have been the motivation for the development of many PMS in the past, there is still no coherent definition what *good* in this context means, i.e. how and what exactly should be seen well. Opinions vary from the implementation of additional viewpoints, strategic links and orientations to perspectives [26]. So, future research on visibility will have to be on the definition of a scale that condenses basic principles of *good* PMS to central requirements and variables that are able to independently assess PMS themselves. Such scale development will be subject to empirical investigation through questionnaires and interviews with the process responsible and stakeholders. Second, the number of cases should be increased to gain a broader empirical basis of applications of the PMS concept, validation and its impact on visibility. In order to do so, we will expand the body of investigation of this paper to the full range of business processes that follow case two. This research will also be on the user acceptance of the approach following the Technology Acceptance Model (TAM).

6. References

[1] H. Schmelzer, W. Sesselmann, "Geschäftsprozess-management in der Praxis", 6. Edition, Hanser, München, 2008, p. 6.

[2] U. Pingel and S. Schmitt, http:// bpmumfrage. wordpress.com, 2007

[3] N. Palmer, "A Survey of Business Process Initiatives in E-Business", BPT Market Survey, BPTrends, 2007.

[4] T. Pidun and C. Felden, "A performance assessment system incorporating indirect indicators and semantics", Proceedings of the Seventeenth Americas Conference on Information Systems, 2011

[5] M. Power "The Audit Society. Rituals of Verification", Oxford University Press, Oxford, 1997.

[6] R. Bose, "Understanding management data systems for enterprise performance management". Industrial Management and Data Systems, Vol. 106, No. 1, Emerald, Bingley, 2006, pp. 43-59.

[7] L. Gonzalez, F. Rubio, F. Gonzalez and M. Velthuis, “Measurement in business processes: a systematic review”, BPMJ, Vol. 16, Issue 1, Emerald, Bingley, 2010, pp.114-134

[8] P. Kueng and T. Wettstein, “Ganzheitliches Performance-Measurement mittels Informationstechnologie”, Haupt, Bern, 2003, p. 44.

[10] M. Genrich, A. Kokkonen, J. Moormann, M. zur Muehlen, R. Tregear, J. Mendling, and B. Weber, “Challenges for Business Process Intelligence: Discussions at the BPI Workshop”, in: A. Ter Hofstede et al. (Eds): Business Process Management Workshops, 2007, Springer, Berlin, 2010

[11] W. Van Der Aalst, H. Reijers, A. Weijters, B. Van Dongen, A. De Medeiros, M. Song, and H. Verbeek, “Business Process Mining: An Industrial Application”, Information Systems, Vol. 32 (5), Elsevier, Amsterdam, 2007, pp. 713-732.

[12] R. Kaplan and D. Norton: Using the Balanced Scorecard as strategic management system, Harvard Business Review January-February 1996, Harvard Business Publishing, Boston, 1996, pp. 75-85.

[13] A. Neely, C. Adams and M. Kennerly: The Performance Prism: The Scorecard for Measuring and Managing Business Success, FT Prentice Hall, London, 2002.

[14] Deloitte Global Services, http://www.deloitte.com/Inthedark, 2007.

[15] P. Harmon, http://www.bptrends.com/publicationfiles/BPT%2520SC%252DSCOR%25202%252D03%252D03%252Epdf.pdf, 2003

[16] J. Wisner and S. Fawcett, “Link firm strategy to operating decisions through performance measurement”, Production and Inventory Management Journal, Third Quarter 1991, APICS, Chicago, 1991, pp.5-11.

[17] J. Zigon, http://www.zigonperf.com/hardstuff.html, 2011.

[18] A. Zerfass, “Rituale der Verifikation? Grundlagen und Grenzen des Kommunikations-Controlling”, in: L. Rademacher (Ed.), Distinktion und Deutungsmacht, VS Verlag für Sozialwissenschaften, Wiesbaden, 2005, pp. 183-222.

[19] D. Ittner and D. Larcker “Coming Up Short on Nonfinancial Performance Measurement”, Harvard Business Review, HBS Publishing, Boston, 2003, pp. 88-95.

[20] P. Bierbusse and T. Siesfeld, “Measures that matter”, Journal of Strategic Performance Measurement, Vol. 1, No. 2, Emerald, Bingley, 1997, pp. 6-11.

[21] M. Bourne, A. Neely, J. Mills and K. Platts, “Implementing performance measurement systems: a literature review”, IJBPM, Vol. 5, No.1, Inderscience, Geneve, 2003, pp. 1-24.

[22] M. Brown, “Keeping Score: Using the Right Metrics to Drive World-class Performance”, Quality Resources, New York, 1996.

[23] P. Kueng, and A. Krahn, “Building a Process Performance Meaasurement System: some early experiences”, Journal of Scientific and Industrial Research, vol. 58, no. 3-4, National Institute of Science Communication and Information Resources, New Delhi, 1999, pp. 149-159.

[24] A. Lönnqvist, “Measurement of Intangible Success Factors: Case Studies on the Design, Implementation and Use of Measures”, Dissertation, Tampere University of Technology, Publication 485, 2004.

[25] D. Larcker, http://www.cimaglobal.com/Documents/Thought_leadership_docs/VisitingProfessor/tech_pres_performance_measures_insight_and_challenges_mar04.pdf, 2004

[26] T. Pidun and C. Felden, “Limitations of Performance Measurement Systems based on Key Performance Indicators”, Proceedings of the Seventeenth Americas Conference on Information Systems, 2011

[27] T. Pidun, J. Buder, and C. Felden, “Optimizing process performance visibility through additional descriptive features in performance measurement”, Proceedings of the Fourth International Workshop on Evolutionary Business Processes (EVL-BP 2011), 2011

[28] M. Caridi, L. Crippa, A. Perego, A. Sianesi and A. Tumino, “Measuring visibility to improve supply chain performance: a quantitative approach”, Benchmarking: An International Journal, Vol. 17, Issue 4, Emerald, Bingley, 2010, pp. 593-615.

[29] A. Hevner, S. Ram and S. March, “Design Science in Information Systems Research”, MIS Quarterly (28)1, Carlson, Minneapolis, 2004, pp. 75-105.

[30] M. Porter and V. Millar, “How information gives you competitive advantage”, Harvard Business Review, July-August, 1985, pp. 149-174.

[31] J. Rüegg-Stürm, “Das neue St. Galler Management-Modell – Grundkategorien einer integrierten Managementlehre”, Haupt, Bern, 2002

[32] G. Zellner, “Leistungsprozesse im Kundenbeziehungs-management”, Dissertation, Universität St. Gallen, 2003

[33] R. Yin, “Case Study Research - Design and Methods”, Third edition, Sage Publications, Beverly Hills, 2003.

[34] A. Borchardt and S. Göthlich, “Erkenntnisgewinn durch Fallstudien”, in: S. Albers et al. (Eds.): “Methodik der empirischen Forschung”, DUV, Wiesbaden 2007, pp.33-48.

[35] J. Creswell, D. Miller, “Determining Validity in Qualitative Inquiry”, Theory into Practice. 39, Nr. 3, Taylor&Francis, Philadelphia, 2000, pp. 124-130.

[36] S. Bhasin “Lean and performance measurement”, Journal of Manufacturing Technology Management, Vol. 19, No. 5, Emerald, Bingley, 2008, pp. 670-684.

[37] N. Senapati “Six Sigma: myths and realities”, International Journal of Quality & Reliability Management, Vol.21, Issue. 6, Emerald, Bingley, 2004, pp. 683–690.

[38] J. Bessant, S. Caffyn, J. Gilbert, R. Harding and S. Webb, “Rediscovering continuous improvement”, Technovation, Volume 14, Issue 1, Elsevier, Amsterdam, 1994, pp. 17-29.

[39] W. Deming, “Out of Crisis”, Cambridge University Press, Cambridge, 1982

[40] V. Popova, A. Sharpanskykh, “Modeling organizational performance indicators”, Information Systems 35(2010), Elsevier, Amsterdam, 2009, p. 505–527.

[41] H. Wei and E. Wang, “The strategic value of supply chain visibility: increasing the ability to reconfigure”, European Journal of Information Systems, Vol. 19, Palgrave, Basingstoke, 2010, pp. 238-249.

[42] T. Davenport, “The coming commoditization of Processes”, Harvard Business Review, June 2005, pp. 101–108.

[43] Deloitte Global Services, http://www.deloitte.com/assets/Dcom-NewZealand/Local%20Assets/Documents/In%20the%20dark%284%29.pdf

[44] J. Broadbent, “If you can’t measure it, how can you manage it”, Public Money and Management, Vol. 27, No.3,Routledge, Abingdon, 2007, pp.193-198.

[45] F. Erhun and P. Keskinocak, “Collaborative Supply Chain Management”, in Planning Production and Inventories in the Extended Enterprise, International Series in Operations Research & Management Science, Volume 151, Springer, Berlin, 2011, pp. 233-268.